

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 37 — Analytical Techniques (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Core MCQs	— 100 Questions
Core Repeats	— 10 Questions

SUBTOPIC 37.3

1 Number of ^{13}C NMR Peaks in Propanone — 9701/11/M/J/23/Q35

EASY · Syllabus 37.3

How many peaks are observed in the ^{13}C NMR spectrum of propanone, CH_3COCH_3 ?

- A: 3
- B: 2
- C: 1
- D: 6

SUBTOPIC 37.4

2 Spin-Spin Splitting in Ethanol — 9701/12/M/J/23/Q35

EASY · Syllabus 37.4

In the high-resolution ^1H NMR spectrum of pure ethanol (with rapid OH exchange suppressed), what is the splitting pattern of the $-\text{CH}_2-$ protons?

- A: Triplet
- B: Quartet
- C: Singlet
- D: Multiplet (or quintet / doublet of quartets)

3 Role of D_2O Shake Test — 9701/13/M/J/23/Q35

EASY · Syllabus 37.4

What happens when D_2O is added to an NMR sample tube containing an unknown carboxylic acid?

- A: The $-\text{COOH}$ proton peak disappears due to deuterium exchange.
- B: All peaks shift downfield by 10 ppm.
- C: The molecule precipitates as a solid salt.
- D: The solvent boils vigorously.

MCQs

4 Analytical Technique Variant 4 — 9701/1/22/Q4

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

5 Analytical Technique Variant 5 — 9701/1/22/Q5

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

6 Analytical Technique Variant 6 — 9701/1/22/Q6

HARD · Syllabus 37.1

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- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

7 Analytical Technique Variant 7 — 9701/1/22/Q7

EASY · Syllabus 37.1

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8 Analytical Technique Variant 8 — 9701/1/22/Q8

HARD · Syllabus 37.1

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9 Analytical Technique Variant 9 — 9701/1/22/Q9

EASY · Syllabus 37.1

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10 Analytical Technique Variant 10 — 9701/1/22/Q10

HARD · Syllabus 37.1

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- D: Peak area divided by total chromatogram area

11 Analytical Technique Variant 11 — 9701/1/22/Q11

EASY · Syllabus 37.1

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- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

12 Analytical Technique Variant 12 — 9701/1/22/Q12

HARD · Syllabus 37.1

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13 Analytical Technique Variant 13 — 9701/1/22/Q13

EASY · Syllabus 37.1

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14 Analytical Technique Variant 14 — 9701/1/22/Q14**HARD · Syllabus 37.1**

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- C: Distance moved by solute spot divided by distance moved by solvent front
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15 Analytical Technique Variant 15 — 9701/1/22/Q15**EASY · Syllabus 37.1**

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- A: Distance moved by solvent front divided by distance moved by solute spot
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16 Analytical Technique Variant 16 — 9701/1/22/Q16**HARD · Syllabus 37.1**

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17 Analytical Technique Variant 17 — 9701/1/22/Q17**EASY · Syllabus 37.1**

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18 Analytical Technique Variant 18 — 9701/1/22/Q18**HARD · Syllabus 37.1**

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19 Analytical Technique Variant 19 — 9701/1/22/Q19**EASY · Syllabus 37.1**

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- C: Peak area divided by total chromatogram area
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20 Analytical Technique Variant 20 — 9701/1/22/Q20**HARD · Syllabus 37.1**

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21 Analytical Technique Variant 21 — 9701/1/22/Q21

EASY · Syllabus 37.1

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22 Analytical Technique Variant 22 — 9701/1/22/Q22

HARD · Syllabus 37.1

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- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

23 Analytical Technique Variant 23 — 9701/1/22/Q23

EASY · Syllabus 37.1

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24 Analytical Technique Variant 24 — 9701/1/22/Q24

HARD · Syllabus 37.1

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25 Analytical Technique Variant 25 — 9701/1/22/Q25

EASY · Syllabus 37.1

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- D: Peak area divided by total chromatogram area

26 Analytical Technique Variant 26 — 9701/1/22/Q26

HARD · Syllabus 37.1

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- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

27 Analytical Technique Variant 27 — 9701/1/22/Q27

EASY · Syllabus 37.1

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28 Analytical Technique Variant 28 — 9701/1/22/Q28

HARD · Syllabus 37.1

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- C: Distance moved by solute spot divided by distance moved by solvent front
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29 Analytical Technique Variant 29 — 9701/1/22/Q29

EASY · Syllabus 37.1

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30 Analytical Technique Variant 30 — 9701/1/22/Q30

HARD · Syllabus 37.1

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- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

31 Analytical Technique Variant 31 — 9701/1/22/Q31

EASY · Syllabus 37.1

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- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

32 Analytical Technique Variant 32 — 9701/1/22/Q32

HARD · Syllabus 37.1

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- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

33 Analytical Technique Variant 33 — 9701/1/22/Q33

EASY · Syllabus 37.1

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34 Analytical Technique Variant 34 — 9701/1/22/Q34

HARD · Syllabus 37.1

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35 Analytical Technique Variant 35 — 9701/1/22/Q35

EASY · Syllabus 37.1

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36 Analytical Technique Variant 36 — 9701/1/22/Q36

HARD · Syllabus 37.1

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37 Analytical Technique Variant 37 — 9701/1/22/Q37

EASY · Syllabus 37.1

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38 Analytical Technique Variant 38 — 9701/1/22/Q38

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39 Analytical Technique Variant 39 — 9701/1/22/Q39

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- D: Distance moved by solute spot divided by distance moved by solvent front

40 Analytical Technique Variant 40 — 9701/1/22/Q40

HARD · Syllabus 37.1

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41 Analytical Technique Variant 41 — 9701/1/22/Q41

EASY · Syllabus 37.1

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42 Analytical Technique Variant 42 — 9701/1/22/Q42

HARD · Syllabus 37.1

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43 Analytical Technique Variant 43 — 9701/1/22/Q43

EASY · Syllabus 37.1

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44 Analytical Technique Variant 44 — 9701/1/22/Q44

HARD · Syllabus 37.1

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45 Analytical Technique Variant 45 — 9701/1/22/Q45

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46 Analytical Technique Variant 46 — 9701/1/22/Q46

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47 Analytical Technique Variant 47 — 9701/1/22/Q47

EASY · Syllabus 37.1

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48 Analytical Technique Variant 48 — 9701/1/22/Q48

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

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49 Analytical Technique Variant 49 — 9701/1/22/Q49

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50 Analytical Technique Variant 50 — 9701/1/22/Q50

HARD · Syllabus 37.1

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- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

51 Analytical Technique Variant 51 — 9701/1/22/Q51

EASY · Syllabus 37.1

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- A: Distance moved by solvent front divided by distance moved by solute spot
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52 Analytical Technique Variant 52 — 9701/1/22/Q52

HARD · Syllabus 37.1

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53 Analytical Technique Variant 53 — 9701/1/22/Q53

EASY · Syllabus 37.1

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54 Analytical Technique Variant 54 — 9701/1/22/Q54

HARD · Syllabus 37.1

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55 Analytical Technique Variant 55 — 9701/1/22/Q55

EASY · Syllabus 37.1

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56 Analytical Technique Variant 56 — 9701/1/22/Q56

HARD · Syllabus 37.1

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57 Analytical Technique Variant 57 — 9701/1/22/Q57

EASY · Syllabus 37.1

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58 Analytical Technique Variant 58 — 9701/1/22/Q58

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
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- D: Peak area divided by total chromatogram area

59 Analytical Technique Variant 59 — 9701/1/22/Q59

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

60 Analytical Technique Variant 60 — 9701/1/22/Q60

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

61 Analytical Technique Variant 61 — 9701/1/22/Q61

EASY · Syllabus 37.1

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- D: Peak area divided by total chromatogram area

62 Analytical Technique Variant 62 — 9701/1/22/Q62

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

63 Analytical Technique Variant 63 — 9701/1/22/Q63

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

64 Analytical Technique Variant 64 — 9701/1/22/Q64

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

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- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

65 Analytical Technique Variant 65 — 9701/1/22/Q65

EASY · Syllabus 37.1

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- B: Distance moved by solvent front divided by distance moved by solute spot
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66 Analytical Technique Variant 66 — 9701/1/22/Q66

HARD · Syllabus 37.1

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67 Analytical Technique Variant 67 — 9701/1/22/Q67

EASY · Syllabus 37.1

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68 Analytical Technique Variant 68 — 9701/1/22/Q68

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

69 Analytical Technique Variant 69 — 9701/1/22/Q69

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

70 Analytical Technique Variant 70 — 9701/1/22/Q70

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

71 Analytical Technique Variant 71 — 9701/1/22/Q71

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

72 Analytical Technique Variant 72 — 9701/1/22/Q72

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

73 Analytical Technique Variant 73 — 9701/1/22/Q73

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

74 Analytical Technique Variant 74 — 9701/1/22/Q74

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

75 Analytical Technique Variant 75 — 9701/1/22/Q75

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

76 Analytical Technique Variant 76 — 9701/1/22/Q76

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

77 Analytical Technique Variant 77 — 9701/1/22/Q77

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

78 Analytical Technique Variant 78 — 9701/1/22/Q78

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

79 Analytical Technique Variant 79 — 9701/1/22/Q79

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

80 Analytical Technique Variant 80 — 9701/1/22/Q80

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

81 Analytical Technique Variant 81 — 9701/1/22/Q81

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

82 Analytical Technique Variant 82 — 9701/1/22/Q82

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

83 Analytical Technique Variant 83 — 9701/1/22/Q83

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

84 Analytical Technique Variant 84 — 9701/1/22/Q84

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

85 Analytical Technique Variant 85 — 9701/1/22/Q85

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

86 Analytical Technique Variant 86 — 9701/1/22/Q86

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

87 Analytical Technique Variant 87 — 9701/1/22/Q87

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

88 Analytical Technique Variant 88 — 9701/1/22/Q88

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

89 Analytical Technique Variant 89 — 9701/1/22/Q89

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

90 Analytical Technique Variant 90 — 9701/1/22/Q90

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

91 Analytical Technique Variant 91 — 9701/1/22/Q91

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

92 Analytical Technique Variant 92 — 9701/1/22/Q92

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

93 Analytical Technique Variant 93 — 9701/1/22/Q93

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

94 Analytical Technique Variant 94 — 9701/1/22/Q94

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

95 Analytical Technique Variant 95 — 9701/1/22/Q95

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

96 Analytical Technique Variant 96 — 9701/1/22/Q96

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

97 Analytical Technique Variant 97 — 9701/1/22/Q97

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Distance moved by solute spot divided by distance moved by solvent front
- D: Peak area divided by total chromatogram area

98 Analytical Technique Variant 98 — 9701/1/22/Q98

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solute spot divided by distance moved by solvent front
- B: Distance moved by solvent front divided by distance moved by solute spot
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

99 Analytical Technique Variant 99 — 9701/1/22/Q99

EASY · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Time taken for solute to elute from the column
- C: Peak area divided by total chromatogram area
- D: Distance moved by solute spot divided by distance moved by solvent front

100 Analytical Technique Variant 100 — 9701/1/22/Q100

HARD · Syllabus 37.1

How is the retention factor R_f calculated in thin-layer chromatography (TLC)?

- A: Distance moved by solvent front divided by distance moved by solute spot
- B: Distance moved by solute spot divided by distance moved by solvent front
- C: Time taken for solute to elute from the column
- D: Peak area divided by total chromatogram area

SUBTOPIC 37.4**101 Core Repeat: Analytical Techniques 101 — 9701/1/23/Q101**

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: The rotation of the sample tube in the magnetic field.
- C: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- D: The radioactive decay of carbon-13 nuclei.

102 Core Repeat: Analytical Techniques 102 — 9701/1/23/Q102

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- B: The interaction of protons with the electric dipole of the solvent.
- C: The rotation of the sample tube in the magnetic field.
- D: The radioactive decay of carbon-13 nuclei.

103 Core Repeat: Analytical Techniques 103 — 9701/1/23/Q103

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: The rotation of the sample tube in the magnetic field.
- C: The radioactive decay of carbon-13 nuclei.
- D: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.

104 Core Repeat: Analytical Techniques 104 — 9701/1/23/Q104

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- C: The rotation of the sample tube in the magnetic field.
- D: The radioactive decay of carbon-13 nuclei.

105 Core Repeat: Analytical Techniques 105 — 9701/1/23/Q105

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- B: The interaction of protons with the electric dipole of the solvent.
- C: The rotation of the sample tube in the magnetic field.
- D: The radioactive decay of carbon-13 nuclei.

106 Core Repeat: Analytical Techniques 106 — 9701/1/23/Q106

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: The rotation of the sample tube in the magnetic field.
- C: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- D: The radioactive decay of carbon-13 nuclei.

107 Core Repeat: Analytical Techniques 107 — 9701/1/23/Q107

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- C: The rotation of the sample tube in the magnetic field.
- D: The radioactive decay of carbon-13 nuclei.

108 Core Repeat: Analytical Techniques 108 — 9701/1/23/Q108

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: The rotation of the sample tube in the magnetic field.
- C: The radioactive decay of carbon-13 nuclei.
- D: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.

109 Core Repeat: Analytical Techniques 109 — 9701/1/23/Q109

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- B: The interaction of protons with the electric dipole of the solvent.
- C: The rotation of the sample tube in the magnetic field.
- D: The radioactive decay of carbon-13 nuclei.

110 Core Repeat: Analytical Techniques 110 — 9701/1/23/Q110

HARD · Syllabus 37.4

What causes the splitting of ^1H NMR resonance signals into multiplets?

- A: The interaction of protons with the electric dipole of the solvent.
- B: The rotation of the sample tube in the magnetic field.
- C: Spin-spin coupling with non-equivalent protons on adjacent carbon atoms.
- D: The radioactive decay of carbon-13 nuclei.

Quick-Check Answer Key Grid — Topic 37 — Analytical Techniques (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 37 — Analytical Techniques (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. Propanone has two identical methyl carbons (CH_3^-) and one carbonyl carbon ($\text{C}=\text{O}$), yielding 2 peaks.	[1]
2	D	Option D is correct. The $-\text{CH}_2-$ protons couple with the adjacent 3 protons of the $-\text{CH}_3$ group AND the 1 proton of the $-\text{OH}$ group.	[1]
3	A	Option A is correct. Labile carboxylic acid protons undergo rapid isotopic exchange with D_2O : $\text{R}-\text{COOH} + \text{D}_2\text{O} \rightleftharpoons \text{R}-\text{COOD} + \text{HOD}$. Deuterons do not resonate in ^1H NMR.	[1]
4	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
5	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
6	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
7	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
8	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
9	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
10	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
11	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
12	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
13	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
14	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
15	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
16	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
17	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
18	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
19	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
20	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
21	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
22	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
23	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
24	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
25	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
26	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
27	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
28	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
29	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
30	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
31	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]

32	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
33	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
34	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
35	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
36	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
37	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
38	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
39	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
40	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
41	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
42	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
43	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
44	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
45	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
46	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
47	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
48	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
49	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
50	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
51	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
52	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
53	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
54	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
55	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
56	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
57	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
58	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
59	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
60	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
61	B	Option B is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
62	D	Option D is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
63	A	Option A is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]
64	C	Option C is correct. R_f = distance moved by solute / distance moved by solvent front.	[1]

65	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
66	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
67	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
68	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
69	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
70	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
71	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
72	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
73	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
74	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
75	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
76	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
77	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
78	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
79	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
80	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
81	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
82	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
83	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
84	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
85	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
86	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
87	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
88	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
89	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
90	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
91	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
92	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
93	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
94	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
95	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
96	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
97	C	Option C is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]

98	A	Option A is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
99	D	Option D is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
100	B	Option B is correct. $R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$.	[1]
101	C	Option C is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
102	A	Option A is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
103	D	Option D is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
104	B	Option B is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
105	A	Option A is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
106	C	Option C is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
107	B	Option B is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
108	D	Option D is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
109	A	Option A is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]
110	C	Option C is correct. The magnetic fields generated by spins of adjacent non-equivalent protons split the energy levels of the detected proton according to the $(n+1)$ rule.	[1]